

SMART INDIA HACKATHON 2026



- **Problem Statement ID –** SIH26039
- **Problem Statement Title-** AI-Powered Underground Mine Safety, Monitoring and Rescue System
- **Theme-** SMART AUTOMATION
- **PS Category-** HARDWARE
- **Team ID-**
- **Team Name-** SAR SQUADS



HANUMA – HAZARD ANALYSIS & NAVIGATION UNIT FOR MINE ASSISTANCE

"HANUMA translates advanced subterranean robotics into a modular, low-cost inspection architecture combining multimodal sensing, resilient communication, and edge intelligence."

- **SUBTERRANEAN HAZARDS: 1. HIGH-RISK ENVIRONMENT**

Underground mines present severe risks including **toxic gases**, **oxygen deficiency**, **dust**, **unstable terrain**, and **communication shadow zones**.

- **THE SOLUTION: HANUMA ROVER**

A rugged **six-wheel rocker-bogie rover** that transforms human-first inspection into a remote, safe, and multimodal sensing operation, designed specifically to traverse and map extreme underground spaces.

CORE INNOVATIONS

1. GPS-DENIED LOCALIZATION

Fusing 2D LiDAR, IMU, and wheel odometry.

2. MULTIMODAL HAZARD INTELLIGENCE

Integrating gas, thermal, vision, and environmental sensors.

3. RESILIENT COMMUNICATION

LoRa-based breadcrumb relay concept.

4. EDGE-AI ASSISTANCE

Local processing for rapid artifact interpretation.

5. FAILSAFE AUTONOMY

Proposed return-to-comms behavior to prevent rover abandonment.



HANUMA HAZARD ANALYSIS & NAVIGATION UNIT FOR MINE ASSISTANCE

Autonomous Six-Wheeled Rover for Underground Mine Inspection, Hazard Detection and Navigation

SENSE • ANALYZE • NAVIGATE • PROTECT
Send the Machine, Not the Human

1. PROBLEM STATEMENT

Underground mines are hazardous and unpredictable environments with:

- Toxic / explosive gases
- Low oxygen
- Poor visibility (dust, smoke)
- Unstable terrain & roof collapse risk
- Inaccessible / narrow passages
- Water accumulation
- Communication limitations
- High risk to human life

HUMAN ENTRY = HIGH RISK

There is a critical need for an autonomous system that can go where humans should not.

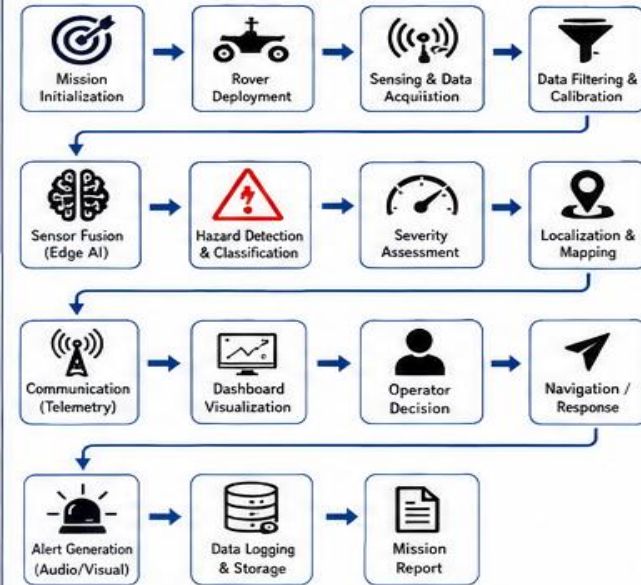
4. EXISTING vs HANUMA

Feature	Existing Systems	HANUMA (Proposed)
Human Operation	Yes (High Risk)	No (Autonomous / Remote)
Gas Detection	Single / Limited	Multi-Gas Real-time
Terrain Mobility	Limited Wheels / Tracks	6WD Rocker-Bogie High Adaptability
Navigation	Manual / No Mapping	Autonomous + Mapping
Hazard Prediction	Not Available	Seismic + AI Prediction
Communication	Short Range / Unreliable	Sub-GHz / LoRa-band Reliable Telemetry
Data Visualization	Basic / No Dashboard	Real-time Dashboard
Edge Intelligence	Not Available	AI-based Sensor Fusion & Decision Support
Localization	Not Available	NavIC/GPS + Mapping
Scalability	Low	High (Modular Design)
Cost	High	Low-Cost Indigenous
Safety	Low	High

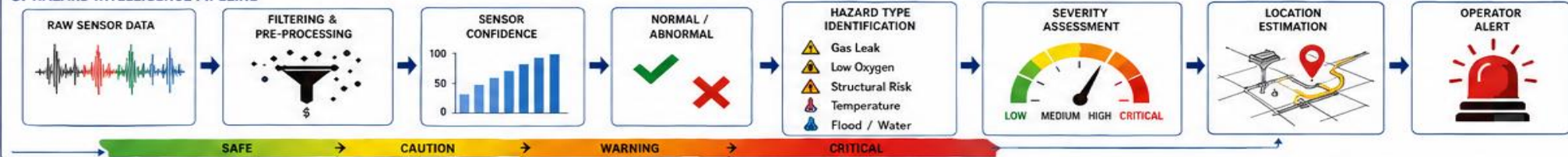
2. SYSTEM ARCHITECTURE



3. WORKFLOW



5. HAZARD INTELLIGENCE PIPELINE

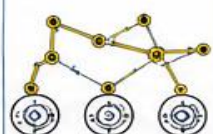


6. ROVER MOBILITY – ROCKER-BOGIE 6WD



High Stability | Maximum Ground Contact | Superior Obstacle Climbing

ROCKER-BOGIE MECHANISM



7. OPERATOR DASHBOARD OVERVIEW



8. KEY FUNCTIONS OF HANUMA



9. SYSTEM SPECIFICATIONS

Dimensions	: 85 x 55 x 45 cm (LxWxH)
Weight	: ~32 kg
Drive System	: 6WD Rocker-Bogie
Max Speed	: 1.5 m/s (approx.)
Battery	: 24V 20Ah Li-ion
Operating Time	: 2 – 3 Hours
Communication	: Sub-GHz / LoRa (1–2 km+)
Protection	: IP67, Dust & Water Proof
Operating Temp	: -10°C to 55°C

AUTONOMOUS INTELLIGENT RUGGED RELIABLE INDIGENOUS

HANUMA enhances mine safety by delivering real-time intelligence, autonomous inspection and hazard mitigation -- saving lives and enabling smarter mining.

APPLICATIONS



HANUMA: Hazard Analysis and Navigation Unit for Mine Assistance

AI-Powered Underground Mine Safety, Monitoring and Rescue System – SIH26039

1. FEASIBILITY ANALYSIS

The proposed system is technically, operationally and economically feasible with current technologies. The architecture leverages proven components and open-source frameworks to ensure a robust, scalable and low-cost solution.

TECHNICAL FEASIBILITY

Subsystem	Technology Used	Feasibility (Level)
 Mobility	Six-Wheel Rocker-Bogie (High Terrain Adaptability)	High
 Sensing Suite	LiDAR, Thermal, Gas & Env. Sensors	High
 Compute	Raspberry Pi 5 + ESP32	High
 SLAM & Navigation	ROS 2, SLAM Toolbox, Nax2	High
 Communication	LoRa Mesh, UHF, 4G/5G (Fallback)	Medium-High
 Power System	Li-ion Battery + BMS	High
 Dashboard	Web-based (ROS 2 + Flask)	High
 AI & Hazard Detection	YOLOv8, Thermal Analysis	Medium-High

OPERATIONAL FEASIBILITY

- ✓ Designed for real underground mine environments (no GPS, low light, dust, gas, variable terrain).
- ✓ Autonomous / semi-autonomous operation reduces human exposure to hazards.
- ✓ Modular payload bay allows mission-specific sensor integration.
- ✓ Easy to deploy and recover using tether/guide rope or remote navigation.

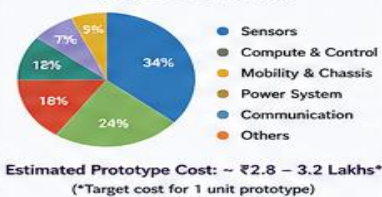
ECONOMIC FEASIBILITY
(Target Cost Analysis)

Fig. 1: Feasibility overview of the HANUMA underground rover system.

FEASIBILITY ANALYSIS, CHALLENGES & RISK MITIGATION STRATEGIES

2. POTENTIAL CHALLENGES AND RISKS

ID	Challenge / Risk	Impact	Likelihood	Risk Level
1	 Loss of Communication (No Wi-Fi / Signal Blockage)	High	High	HIGH
2	 Dust, Smoke and Poor Visibility Affecting Sensors	High	High	HIGH
3	 Toxic / Flammable Gas Presence (CH ₄ , CO, H ₂ S etc.)	High	Medium	HIGH
4	 Extreme Temperature & Humidity Variations	Medium	Medium	MEDIUM
5	 Tunnel Collapse / Falling Debris / Dynamic Obstacles	High	Medium	HIGH
6	 Rough Terrain, Water, Mud and Slippery Surfaces	High	High	HIGH
7	 Power Limitations / Battery Degradation	Medium	Medium	MEDIUM
8	 Localization Drift / SLAM Failure in Feature-Poor Areas	High	Medium	HIGH
9	 RF Interference / Signal Jamming	Medium	Low	LOW
10	 Water Ingress / Flooding	High	Low	MEDIUM

Risk Level: ● HIGH ● MEDIUM ● LOW

ASSUMPTIONS

- Environment similar to Indian underground coal mines (Jharia/Dhanbad belt).
- Rover operates in GPS-denied, low-visibility, high-risk conditions.
- Communication infrastructure is limited and non-uniform.
- Human supervision available at the surface control station.

KEY FEASIBILITY SUMMARY



3. STRATEGIES FOR OVERCOMING CHALLENGES



Communication Loss

- Use LoRa Mesh for robust multi-hop communication.
- Auto-switch to UHF (ground-penetrating) fallback.
- Store-and-Forward data buffering when link is down.



Poor Visibility (Dust/Smoke)

- Combine LiDAR + Thermal + Inertial sensing.
- Apply sensor fusion and adaptive filtering.
- Use IR illumination and thermal imaging.



Hazardous Gas Presence

- Real-time gas monitoring with multi-gas sensors.
- AI-based early hazard detection & trend analysis.
- Automatic alerts and evacuation assistance.



Temperature & Humidity Extremes

- Use industrial-grade components.
- Thermal throttling and ventilation design.
- IP65 enclosure for environmental protection.



Tunnel Collapses / Debris

- 6WD rocker-bogie for stability and obstacle clearance.
- LiDAR-based 2D mapping for obstacle detection.
- Safe path planning with obstacle avoidance.



Rough / Slippery Terrain

- High-torque motors with closed-loop control.
- Traction-optimized tires and rocker-bogie suspension.
- Real-time slip detection and adaptive control.



Power Limitations

- High-capacity Li-ion battery with BMS.
- Power optimization & adaptive mission planning.
- Hot-swappable battery (future upgrade).



Localization Drift / SLAM Failure

- LiDAR + IMU fusion (TILIS SLAM).
- Loop closure detection and map refinement.
- Re-localization using visual / LiDAR features.



RF Interference / Jamming

- Spread spectrum (LoRa) & frequency hopping.
- Mesh routing and redundant channels.
- Interference-resilient communication protocols.



Water Ingress / Flooding

- IP65 sealed chassis and connectors.
- Drainage design & corrosion-resistant materials.
- Water level sensing and auto mission adaptation.

Fig. 2: Analysis of feasibility, potential challenges and risk mitigation strategies for HANUMA underground mine safety and rescue system.

IMPACT ON TARGET AUDIENCE & BENEFITS OF THE SOLUTION

1. TARGET AUDIENCE

**Mine Workers**

Ensures safer working conditions and reduces exposure to underground hazards.

**Rescue Teams & Safety Officers**

Provides real-time situational awareness for faster, informed and safer decision-making.

**Mine Management**

Improves operational efficiency, compliance and risk management.

**Government & Regulatory Bodies**

Supports enforcement, monitoring and improvement of mine safety standards.

**Local Communities & Society**

Reduces accidents, saves lives and enhances trust in mining operations.

2. POTENTIAL IMPACT OVERALL



3. BENEFITS OF THE SOLUTION

**SOCIAL BENEFITS**

- Saves lives and prevents injuries.
- Enhances well-being of mine workers.
- Strengthens community trust and social acceptance of mining.

**ECONOMIC BENEFITS**

- Reduces economic losses due to accidents, downtime and equipment damage.
- Lowers operation and inspection costs.
- Improves productivity and operational efficiency.
- Enables cost-effective remote inspections.

**ENVIRONMENTAL BENEFITS**

- Helps detect gas leaks and prevent environmental contamination.
- Supports responsible and sustainable mining operations.
- Reduces resource wastage through efficient monitoring.

**TECHNOLOGICAL & STRATEGIC BENEFITS**

- Strengthens India's capabilities in underground robotics and AI.
- Encourages innovation and self-reliance.
- Supports adoption of Industry 4.0 technologies in mining.

4. IMPACT IN KEY SCENARIOS

Gas Leak / Toxic Hazard

Early detection of gases (CH₄, CO, CO₂, H₂S etc.) prevents explosions and suffocation.

Tunnel Collapse / Debris

Helps identify safe paths, assess stability and locate trapped individuals.

Fire / High Temperature

Thermal imaging helps detect hotspots and spread, enabling timely firefighting response.

Flooding / Water Ingress

Water level and humidity monitoring enables early flood detection and warning.

Low Visibility / Smoke

LiDAR and thermal sensors ensure navigation and situational awareness in zero-visibility.

5. LONG-TERM VISION & SUSTAINABLE OUTCOMES

**Safe & Resilient Mines**

Safer underground operations with minimal human risk.

**Skilled Workforce & Training**

Data-driven training and capacity building for better preparedness.

**Innovation & Research**

Creates a foundation for future autonomous underground systems.

**Sustainable Mining**

Environment-friendly and responsible mining practices.

**Stronger Society & Economy**

Contributes to national safety, productivity and economic growth.

HANUMA aims to make underground mining safer, smarter and more sustainable through intelligent automation and real-time awareness.

Technology for Safety. Innovation for Humanity.

RESEARCH AND REFERENCES

CERBERUS in the DARPA Subterranean Challenge (Science Robotics, 2022):

<https://www.science.org/doi/10.1126/scirobotics.abp9742>

DARPA Subterranean (SubT) Challenge Official Overview:

<https://www.darpa.mil/program/darpa-subterranean-challenge>

Directorate General of Mines Safety (DGMS) - Official Statistics & Circulars:

<https://dgms.gov.in/>

ROS 2 Navigation2 (Nav2) Official Documentation:

<https://navigation.ros.org/>

OUTPUTS:- <https://drive.google.com/drive/folders/1AN-MhGUkb1pW22zvBqY6sv8BgVrvYtHC?usp=sharing>

""The above given link contains the outputs obtained from the testing of our original model/prototype.""

GITHUB:- https://github.com/Gopinayak02/HANUMA_Weekly_Report

The Pipelines, we went through our model processing, and we also add the training .pth files to avail the trained weights directly.

NOTE:- The model weights are under **re-enhancing stage** which will be released on and before i.e., 27/10/2026.

YOUTUBE LINK:- https://youtube.com/shorts/cCB2XQQ_4Ew?si=7JsAbaE7Y5XIR80W

